R37-01 radiation and STF5 shear sensitivity

The positive synthetic R36 a0.02_b0.3 temperature sky and saved paired R36 noise draws determine this conditional methods image. The same sky, local observer, nuisance, correlated eight-coordinate temperature law, and constructed central derivative target are used for every computable adapter. The derivative is an **assumption**, not a measured radiation history.

Full direct quadrature and exact (rho, pi, H4) common-carrier response differ by 8.88e-16 in normalized energy units, below the preregistered 2e-10 tolerance. The full response eigenvalues are [0.539469 0.539473 0.637188 0.638056 0.664453] and its condition number is 1.2317. The generic joint gain/shear carrier has rank 23, or 20 at V=0, as reused from Goal 1. For this fixed screen profile the actual angular kernels have ranks 15 (full), 11 (H4-dropped with separate rho/pi/P_star), and 6 (true V=0 Thomson gain M). The screen remains positive with minimum physical eigenvalue 0.41555.

The H4-dropped closure retains separate rho, pi and P_star; its central STF5 bias norm is 4.7577e-05. It is not a gain-only estimator. At fixed R_tau and fixed screen profile, two strictly positive brightness fields have the same true rank-6 gain M to 1.11e-15 yet their true shear solutions differ in norm by 0.000118028. Thus M alone identifies a set, not one shear. This deterministic positive fibre witness is separate from the retained R36 stochastic draw law. GAIN_IDENTIFIED_SUPPORT.csv gives a two-member inner witness of this set and its common-disk physical sensitivity; the full positive M-fibre may be wider. Omitting P_star with the exact intensity response gives full-minus-unpolarized bias norm 0.0085038, bounded by gamma_T ||P_star||/(5 lambda_min) = 0.0087824.

One shared statistical event uses the maximum standardized centered shear deviation over all 20 mode-coordinate pairs on the retained 8191 calibration draws. Its higher-quantile threshold is 1.27089; simultaneous centered test coverage on the same 4096 saved draws is 3888/4096 = 0.949219. The earlier five marginal full-response intervals jointly hit only 3389/4096, retained as a regression diagnostic. MATCHED_COVERAGE.csv labels marginal diagnostics separately and reports joint radii, widths, target coverage and signed bias. The common event includes saved-draw operator variation because each response is recomputed on each draw.

The physical sensitivity uses a single two-coordinate unit-disk primitive z. The same z drives derivative and polarization errors through

the displayed U,V matrices; their effects are summed before one central inverse map. SHEAR_SUPPORT.csv gives fixed-operator directional support over the declared budgets. Its triangle half-width is a sensitivity layer; combined physical/statistical coverage has not been calibrated. Structural ranks and five-trial median inference-only timings on identical precomputed test arrays appear in JSON; timings exclude carrier extraction.

The figures and tables have owner htt, claim tier conditional_synthetic_method_experiment, artifact mode finite_methods_image, transfer source none, synthetic null status and the same R36 correlated noise law. Allowed use is a conditional response/shear sensitivity image. Empirical shear, native solver validation and Bianchi family identification are forbidden uses. Original 42-node acceptance remains 17/42; analytic alternate remains 18/42.